PROJECT ROLES AND VIVA PREPARATION GUIDE

Project Title: Network Anomaly Detection Pipeline using Machine Learning **Team:** Subhrajit Ghorai, Jullie Mahato, Purbasha Santra

1. INDIVIDUAL ROLES & RESPONSIBILITIES

In a viva, examiners often ask: *"What was your specific contribution?"* Use these tailored answers:

Subhrajit Ghorai (Lead Developer & ML Architecture)

- **Responsibility:** Model implementation and System Orchestration.
- **Key Contribution:** "I developed the `model_trainer.py` and `main.py` scripts. I implemented the three different algorithms (Random Forest, MLP, and Isolation Forest) and handled the logic for model evaluation and the generation of Confusion Matrices."

Jullie Mahato (Data Handler & Pre-processing)

- **Responsibility:** Data Ingestion and Feature Engineering.
- **Key Contribution:** "I focused on the `data_handler.py` script. I was responsible for cleaning the KDD dataset, specifically handling the 'trailing periods' in labels, and implementing One-Hot Encoding to convert categorical network protocols into numerical data for the models."

Purbasha Santra (Documentation & Performance Analyst)

- **Responsibility:** Technical Writing and Result Interpretation.
- **Key Contribution:** "I managed the project documentation and analyzed the experimental results. I interpreted the classification reports and identified why the Random Forest model outperformed the others, ensuring our findings were aligned with Cyber Forensics standards."

2. CORE PROJECT TOPICS (The "Quick Summary")

What is an Anomaly?

An anomaly is any network traffic that deviates from "Normal" behavior. It could be a DoS (Denial of Service), a Probe (scanning ports), or unauthorized access attempts.

The Three-Model Approach

- **Random Forest (Supervised):** Our best performer. It uses an ensemble of trees to make a majority-vote decision.
- **Keras MLP (Deep Learning):** A neural network that mimics the human brain to find complex, non-linear patterns.

- **Isolation Forest (Unsupervised):** This is for "Zero-Day" attacks. It finds anomalies by isolating them from the rest of the data without needing to know the "answers" beforehand.

The Pre-processing Pipeline

Raw network data is messy. We used **One-Hot Encoding** because models can't "read" words like `TCP` or `HTTP`. We turn those into `1s` and `0s`.

3. TOP 10 LIKELY VIVA QUESTIONS & ANSWERS

Q1: Why is Random Forest getting 1.00 (100%) accuracy?

- **Answer:** "The KDD dataset has very distinct patterns in its 'service' and 'flag' features. After our One-Hot Encoding, these features become highly separable, allowing the Random Forest to create perfect decision boundaries."

Q2: What is the difference between Supervised and Unsupervised learning in your project?

- **Answer:** "Random Forest and MLP are **Supervised**—they learn from labeled data (Normal vs. Attack). Isolation Forest is **Unsupervised**—it finds anomalies purely by looking for data points that are statistically different or 'isolated' from the majority."

Q3: What is a Confusion Matrix?

- **Answer:** "It's a table used to describe the performance of a classification model. It shows True Positives (correctly identified attacks) and False Positives (normal traffic wrongly flagged as an attack)."

Q4: Which is more dangerous: A False Positive or a False Negative?

- **Answer:** "In Cyber Forensics, a **False Negative** is much more dangerous. It means an actual attack happened but the system missed it. A False Positive is just a 'false alarm'."

Q5: What is "Overfitting"?

- **Answer:** "It's when a model learns the training data *too* well, including the noise, and fails on new data. We used a **Dropout Layer** (0.2) in our Keras MLP to prevent this."

Q6: Why did you use the KDD Cup 99 dataset?

- **Answer:** "It is the world-recognized benchmark for testing network intrusion detection systems. It provides 41 different features of a network connection, which is ideal for behavioral analysis."

Q7: How did you handle categorical data?

- **Answer:** "We used `pd.get_dummies()` for One-Hot Encoding. This converts columns like 'Protocol Type' (TCP, UDP, ICMP) into separate binary columns so the machine can process the math."

Q8: What is the "Contamination Factor" in Isolation Forest?

- **Answer:** "It represents the percentage of anomalies we expect in the data. Based on our training set, we set it to approximately 0.46 (46%)."

Q9: What is the "Future Scope" of this project?

- **Answer:** "Moving from 'Batch Processing' (CSV files) to 'Real-Time' detection. We could use the **Scapy** library to sniff live network packets and feed them into our trained model instantly."

Q10: How does this project relate to Cyber Forensics?

- **Answer:** "It automates the 'Detection' phase of forensics. Instead of an investigator manually looking through millions of log entries, our system flags the top 1% of suspicious traffic for them to examine."

4. VIVA TIPS

1. **Don't panic if you don't know a specific code line:** Focus on the *logic*. Explain *why* you used the library (e.g., "We used Pandas for data cleaning").
2. **Point to the charts:** If you have the Confusion Matrix or Feature Importance plots open, refer to them. Examiners love visuals.
3. **Mention the Teamwork:** Use "We" when talking about the project goals, and "I" when talking about your specific role listed above.